Table A.6: Gene-environment interplay: Long-term effects of the measles vaccine introduction – PGIs based on a UK Biobank GWAS

	Years of education				Height in cm			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post-vaccine share 1 to 6 × Pre-vacc. measles	0.225***	0.199***	0.034	−0.023	2.054***	2.046***	0.348	0.086
	(0.025)	(0.022)	(0.101)	(0.104)	(0.061)	(0.054)	(0.221)	(0.222)
PGI	0.552***	0.487***	0.488***	0.487***	3.096***	3.061***	3.065***	3.064***
	(0.009)	(0.007)	(0.007)	(0.007)	(0.015)	(0.015)	(0.015)	(0.015)
Post-vaccine share 1 to 6 × Pre-vacc. measles × PGI	−0.009	−0.013	−0.014	−0.014	0.077*	0.082*	0.073	0.087**
	(0.019)	(0.019)	(0.019)	(0.020)	(0.044)	(0.044)	(0.045)	(0.045)
Controls for:								
Gender	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month of birth FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
District of birth FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
School year of birth FE	No	No	Yes	Yes	No	No	Yes	Yes
County-specific birthdate trend	No	No	No	Yes	No	No	No	Yes
Compulsory schooling 16	Yes	Yes	No	No	No	No	No	No
Compulsory schooling 16 × Pre-vacc. measles	Yes	Yes	Yes	Yes	No	No	No	No
Comp. schooling 16 × Pre-vacc. measles × PGI	Yes	Yes	Yes	Yes	No	No	No	No
Principal components 1-20	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	170,158	170,134	170,134	170,134	170,750	170,725	170,725	170,725

Note: The explanatory variable of interest is the share of the age period from 1 to 6 years during which the individual was exposed to the vaccination program, interacted with the measles cases per 100 people prior to the vaccination program. This measure of treatment intensity for the vaccine introduction is furthermore interacted with the polygenic index (PGI) for education (columns 1-4) / height (columns 5-8). The PGIs were constructed using the summary statistics from a GWAS on individuals from the UK Biobank (excluding the sample used for our main analysis). Individuals born in districts that participated in the 1966 trial are excluded from the samples. Standard errors clustered at the district of birth level are shown in parentheses. Significance levels are indicated as follows: * p<0.1, ** p<0.05, *** p<0.01

Table A.7: Gene-environment interplay in the sibling sample: Long-term effects of the measles vaccine introduction

	Years of education				Height in cm			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post-vaccine share 1 to 6 × Pre-vacc. measles	0.228***	0.190**	0.190	0.124	1.891***	2.176***	2.500**	2.203*
	(0.085)	(0.085)	(0.415)	(0.433)	(0.224)	(0.220)	(1.085)	(1.265)
PGI	0.651***	0.592***	0.591***	0.591***	3.097***	3.100***	3.102***	3.092***
	(0.030)	(0.033)	(0.032)	(0.033)	(0.073)	(0.077)	(0.077)	(0.078)
Post-vacc. share 1 to 6 × Pre-vacc. measles × PGI	0.000	−0.018	−0.013	−0.013	0.114	0.130	0.113	0.155
	(0.086)	(0.089)	(0.090)	(0.093)	(0.231)	(0.251)	(0.252)	(0.258)
Controls for:								
Gender	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month of birth FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
District of birth FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
School year of birth FE	No	No	Yes	Yes	No	No	Yes	Yes
County-specific birthdate trend	No	No	No	Yes	No	No	No	Yes
Compulsory schooling 16	Yes	Yes	No	No	No	No	No	No
Compulsory schooling 16 × Pre-vacc. measles	Yes	Yes	Yes	Yes	No	No	No	No
Comp. schooling 16 × Pre-vacc. measles × PGI	Yes	Yes	Yes	Yes	No	No	No	No
Principal components 1-20	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	10,763	10,553	10,553	10,553	10,803	10,592	10,592	10,592

Note: The sample was restricted to full siblings - identified based on genetic relatedness. The explanatory variable of interest is the share of the age period from 1 to 6 years during which the individual was exposed to the vaccination program, interacted with the measles cases per 100 people prior to the vaccination program. This measure of treatment intensity for the vaccine introduction is furthermore interacted with the polygenic index (PGI) for education (columns 1-4) / height (columns 5-8). The measure of genetic propensity for education is based on summary statistics from Lee *et al.* (2018), the measure for height is based on summary statistics from Wood *et al.* (2014). Individuals born in districts that participated in the 1966 trial are excluded from the samples. Standard errors clustered at the district of birth level are shown in parentheses. Significance levels are indicated as follows: * p<0.1, ** p<0.05, *** p<0.01